

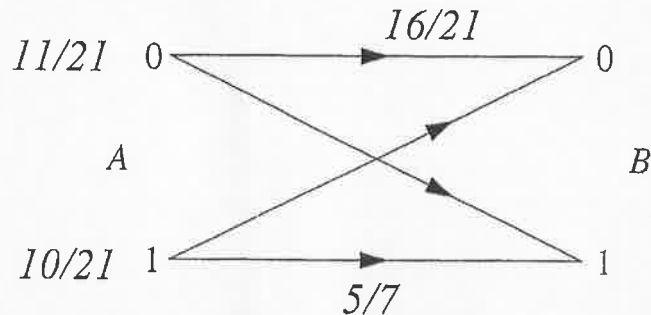


Name of the Examination: WINTER 2022-2023 – CAT-1

School : School of Electronics Engineering
Course Code : CSE2013 Course Title : Information Theory and Coding
Set Number : 01 Date of exam : 14-02-2023 (FN)
Duration : 90 Min Total Marks : 50 (BI)

Answer all the questions

- Q1** A PCM (pulse-coded modulated) source transmits four samples (a_1, a_2, a_3 & a_4) with a rate 25 kilo symbols per second. The probabilities of the symbols are $p(a_1) = p(a_3) = p(a_5) = 2/9$ and $p(a_2) = p(a_4) = p(a_6) = 1/9$. Find the information rate of the source. **8**
- Q2** Consider the following binary channel:



Calculate $H(A)$, $H(B)$, $H(A|B)$, $H(B|A)$, $H(A, B)$ and $I(A; B)$

- Q3** A submarine is sending messages using Morse codes (dash “-” and dot “.”) to the headquarters (HQ). As the submarine is in deep sea, 10% percent of dash was received as dot and 10% of dot was received as dash at the HQ. If the transmission of dot and dash are equiprobable, what is the amount of uncertainty resolved about the transmitted messages after the HQ has observed the received messages. **12**
- Q4** Two sources whose outcomes belong to same source alphabet with different probability distributions are given as $P = [0.2, 0.1, 0.2, 0.35, 0.15]$ and $Q = [0.27, 0.25, 0.15, 0.23, 0.1]$. Find the relative entropies (i) $D(P, Q)$ and (ii) $D(Q, P)$ **10**
- Q5** Let X and Y are two independent random variables with probabilities $P(X) = \{0.25, 0.45, 0.3\}$ and $P(Y) = \{0.15, 0.5, 0.35\}$. Find the joint entropy $H(X, Y)$. **8**

QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	1	CO1	PO2	PEO2	PSO2	8
Q2	1	CO1	PO1	PEO1	PSO1	12
Q3	1	CO1	PO1	PEO1	PSO3	12
Q4	1	CO1	PO2	PEO1	PSO1	10
Q5	1	CO1	PO1	PEO1	PSO1	8

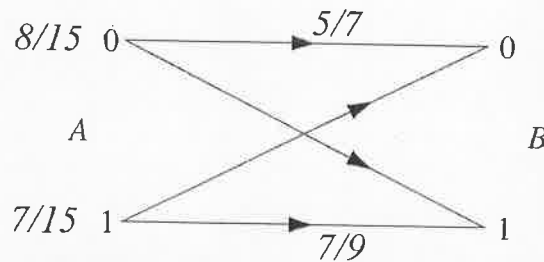
Name of the Examination: WINTER 2022-2023 – CAT-1

School : School of Electronics Engineering
 Course Code : CSE2013 Course Title : Information Theory and Coding
 Set Number : 02 Date of exam : 13-02-2023 (AI)
 Duration : 90 Min Total Marks : 50 (FN)

Answer all the questions

- Q1** An analog-to-digital converter (ADC) converts a 4kHz band limited signal into digital signal at a rate equals to Nyquist rate. The digital signal has 6 quantization levels with probabilities $P = \{0.2, 0.15, 0.1, 0.3, 0.25\}$. What is the information rate given by the ADC. **8**

- Q2** Consider the following binary channel: **12**



Calculate $H(A)$, $H(B)$, $H(A|B)$, $H(B|A)$, $H(A, B)$ and $I(A; B)$

- Q3** Two noisy channels XY and YX are cascaded with their channel matrices given below. The input probabilities are given as $P(x_1) = P(x_2) = \frac{1}{2}$. Find the overall mutual information $I(X; Z)$ **12**

$$P\left(\frac{y}{x}\right) = \begin{bmatrix} \frac{2}{3} & \frac{1}{6} & \frac{1}{6} \\ \frac{1}{7} & \frac{1}{7} & \frac{5}{7} \end{bmatrix}, P\left(\frac{z}{y}\right) = \begin{bmatrix} \frac{5}{8} & \frac{1}{8} & \frac{1}{4} \\ \frac{1}{9} & \frac{2}{3} & \frac{2}{9} \\ \frac{1}{8} & \frac{1}{8} & \frac{3}{4} \end{bmatrix}$$

- Q4** Let a source generates the symbols $\mathcal{A} = \{a_1, a_2, a_3, a_4\}$ with probabilities $\{0.35, 0.25, 0.1, 0.3\}$ respectively. Find the information content (in bits) present in the messages (i) $a_2 a_4 a_1$ (ii) $a_1 a_2 a_3$ (iii) $a_4 a_3 a_2$ (iv) $a_1 a_4 a_3$ (v) $a_1 a_2 a_2$ **10**
- Q5** Let X and Y are two **independent** random variables with probabilities $P(X) = \{0.45, 0.3, 0.25\}$ and $P(Y) = \{0.35, 0.5, 0.15\}$. Find the conditional entropy $H(X, Y)$. **8**

QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	1	CO1	PO2	PEO2	PSO2	8
Q2	1	CO1	PO1	PEO1	PSO1	12
Q3	1	CO1	PO1	PEO1	PSO3	12
Q4	1	CO1	PO2	PEO1	PSO1	10
Q5	1	CO1	PO1	PEO1	PSO1	8